

APPENDIX

TABLE I

EXPLANATION OF THE SYMBOLS USED IN THE PAPER.

Used Symbols	Descriptions
CAS	Central Aggregation Server
EOCL	"Ensemble of Classifiers" with m classifiers either in series or in parallel
U and τ	Global and local iterations respectively
$\mathcal{E}$	Globally trained EOCL
G	Globally trained learning model
$C_i: i = 1 \dots n$	Local edge clients
$\mathcal{E}_i: i = 1 \dots n$	Local replica of global EOCL at C_i 's
$G_i: i = 1 \dots n$	Local replica of G
$\mathcal{B}_i^t: i = 1 \dots n$	Private data of active C_i
$\mathcal{B}_i^t: i = 1 \dots n$	Filtered (using local EOCL) private data of active C_i
$Cl_j: j = 1 \dots N$	Classes in the multi-modal training data
$Cl_j M_i: j = 1 \dots N; i = 1 \dots m$	i -th classifier model trained on the i -th partition of Cl_j
$\mathcal{T}$	Pristine training set
$\mathcal{T}^\circ$	Poisoned training set
$\delta \alpha_j$	Accuracy metric for class j or Cl_j
$E(C_i) = \{D_i^t, D_i^u\}$	Execution environment in active edge C_i
D_i^t and D_i^u	Trusted enclave and untrusted domain in C_i
H_i^{BIN} and H_i^{CFG}	Hash of executable binaries, and reference hash of approved configuration runtime respectively in C_i
$\mathcal{A}$	A set potential adversaries
$\nabla_{\theta_i} G_i$	Local gradient in C_i
$\mathcal{H}$	A set of potential active honest edges
iid	Independent and Identically Distributed, i.e., Data samples are independent and drawn from the same distribution
$\beta \leq 1$	Poison fraction, i.e., fraction of clients among the total number of active edges are corrupted

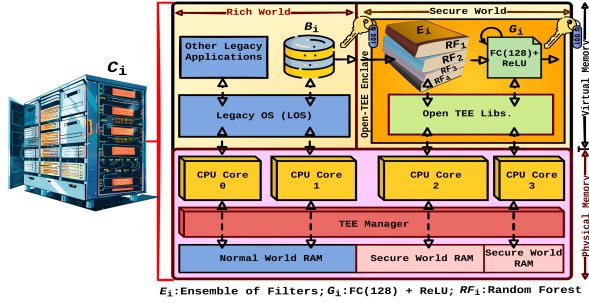


Fig. 1. Overview of the secure enclave in any active edge. Here, $\mathcal{E}_i$ contains four RFs with different parameter settings and the local G_i as FC(128)+ReLU.

A. Proof of Theorem 1

Let us partition the active edges $\{C_i\}_{i=1}^M$ into honest ($\mathcal{H}$, with $|\mathcal{H}| = (1 - \beta)M$) and adversarial ($\mathcal{A} = \beta M$) subsets. Then, $\theta_{\text{GLOBAL}}^{t+1} = (1 - \beta)\hat{\theta}_{\mathcal{H}}^{t+1} + \beta\hat{\theta}_{\mathcal{A}}^{t+1}$, where $\hat{\theta}_{\mathcal{H}}^{t+1} = \frac{1}{(1-\beta)M} \sum_{i \in \mathcal{H}} w_i \mathcal{E}_i(\theta_i^{t+1})$ and analogously for $\hat{\theta}_{\mathcal{A}}^{t+1}$. The clean attack-free aggregation is $\theta_{\text{CLEAN}}^{t+1} = \hat{\theta}_{\mathcal{H}}^{t+1}$.

Subtracting we get

$$\|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\|_2 = \beta \|\hat{\theta}_{\mathcal{A}}^{t+1} - \hat{\theta}_{\mathcal{H}}^{t+1}\|. \quad (1)$$

For any adversarial client $i \in \mathcal{A}$,

$$\begin{aligned} \|\mathcal{E}_i(\theta_i^{t+1}) - \hat{\theta}_i^{t+1}\|_2 &\leq \|\theta_i^{t+1} - \hat{\theta}_i^{t+1}\|_2 + \|\mathcal{E}_i(\theta_i^{t+1}) - \theta_i^{t+1}\|_2 \\ &\leq \Delta + \delta \alpha_i \leq 2\Delta \end{aligned} \quad (2)$$

Therefore, by Jensen's inequality for convex norms,

$$\|\hat{\theta}_{\mathcal{A}}^{t+1} - \hat{\theta}_{\mathcal{H}}^{t+1}\|_2 \leq 2\Delta \quad (3)$$

Substituting Eq. 3 in to Eq. 1 we get the following:

$$\|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\|_2 \leq 2\beta\Delta. \quad (4)$$

TABLE II

PARAMETER SETTINGS DETAILS (FOR REPRODUCIBILITY) OF THE EOCLS USED IN THE EXPERIMENTS. NOTE: ALL PARAMETERS ARE SCIKIT-LEARN V1.6.1 DEFAULTS UNLESS BOLDED; BOLD = VALUES EXPLICITLY SET IN OUR RUNS.

Parameter	Preprocess	SVM	RF	MLP
with_mean	True	—	—	True
with_std	True	—	—	True
copy	True	—	—	True
C	—	4.0	—	—
kernel	—	rbf	—	—
gamma	—	scale	—	—
degree	—	3	—	—
coef0	—	0.0	—	—
shrinking	—	True	—	—
probability	—	False	—	—
tol	—	1e-3	—	1e-4
cache_size	—	200	—	—
class_weight	—	None	—	—
verbose	—	False	0	False
max_iter	—	-1	—	30
decision_function_shape	—	ovr	—	—
break_ties	—	False	—	—
random_state	—	42	42	42
n_estimators	—	—	200	—
criterion	—	—	gini	—
max_depth	—	—	None	—
min_samples_split	—	—	2	—
min_samples_leaf	—	—	1	—
min_weight_fraction_leaf	—	—	0.0	—
max_features	—	—	sqrt	—
max_leaf_nodes	—	—	None	—
min_impurity_decrease	—	—	0.0	—
bootstrap	—	—	True	—
oob_score	—	—	False	—
n_jobs	—	—	-1	—
warm_start	—	—	False	False
ccp_alpha	—	—	0.0	—
max_samples	—	—	None	—
hidden_layer_sizes	—	—	—	(256, 128)
activation	—	—	—	relu
solver	—	—	—	adam
alpha	—	—	—	1e-4
batch_size	—	—	—	256
learning_rate	—	—	—	constant
learning_rate_init	—	—	—	0.001
power_t	—	—	—	0.5
shuffle	—	—	—	True
early_stopping	—	—	—	True
validation_fraction	—	—	—	0.1
momentum	—	—	—	0.9
nesterovs_momentum	—	—	—	True
beta_1	—	—	—	0.9
beta_2	—	—	—	0.999
epsilon	—	—	—	1e-8
n_iter_no_change	—	—	—	3
max_fun	—	—	—	15000

For weighted averaging with normalization $\frac{1}{1-\beta}$ to preserve unbiasedness, the tight constant becomes $\frac{2\beta}{1-\beta}$, giving the final bound. If the global objective $\mathcal{U}(\theta)$ is L -smooth,

$$\begin{aligned} |\mathcal{U}(\theta_{\text{GLOBAL}}^{t+1}) - \mathcal{U}(\theta_{\text{CLEAN}}^{t+1})| &\leq L \|\theta_{\text{GLOBAL}}^{t+1} - \theta_{\text{CLEAN}}^{t+1}\|_2 \\ &\leq \frac{2L\beta}{1-\beta} \Delta, \end{aligned} \quad (5)$$

establishing bounded loss degradation and stability of subsequent updates.

1) Interpretation & Insight: (1) *Admissibility as trust region:* The filter $\mathcal{E}_i$ projects each update into a convex “safe zone” of radius Δ around the honest manifold; no local update can pull the global model outside this region; (2) *Linear robustness law:* Global deviation grows linearly with attack fraction β and local divergence Δ , giving a measurable security budget: halving Δ or β halves the deviation; (3) *Tightness:* The constant $\frac{1}{1-\beta}$ is tight for affine aggregation under worst-case poisoning; stronger filters (e.g., trimmed-mean EOCL) yield smaller empirical constants; and (4) *System relevance:* The bound translates to a Zero-Trust contract: if every client attestation ensures Δ -bounded admissibility, then the entire FL process preserves integrity within an analytically provable envelope.

B. Proof of Theorem 2

The execution environment $E(C_i)$ in an active edge C_i is defined as $E(C_i) = \{D_i^u, D_i^t\}$, where D_i^u and D_i^t , respectively, the untrusted (fully observable and controllable by adversary $\mathcal{A}$) and trusted environments in an active edge C_i . The local secret tuple $\mathcal{S}_i = \{\theta_i, \nabla_{\theta_i} G_i, \mathcal{E}_i\}$ resides entirely within D_i^t . The trusted domain runs a measured boot chain: $\text{CAS} \rightarrow \text{ATTESTATION } D_i^t$, producing a signed quote $\sigma_i = \text{Sign}_{\text{CAS}}(H(D_i^t))$. Only if $H(D_i^t)$ matches the authorized policy manifest $\mathcal{P}_i$ does the enclave receive its session keys and policy tokens.

1) *Confidentiality Argument*: Let K_i be the symmetric encryption key provisioned by CAS after attestation. All enclave state variables— $\theta_i, \nabla_{\theta_i} G_i$ and classifier ensemble $\mathcal{E}_i$ are stored and transmitted as $C_i = \text{Enc}_{K_i}(\mathcal{S}_i || r)$, where r is nonce-based randomness. Even if $\mathcal{A}$ intercepts or copies C_i through DMA, I/O, or host memory inspection, it cannot recover $\mathcal{S}_i$ without K_i , since $\mathbb{P}[\text{Dec}_{K_i}(c_i) = \mathcal{S}_i] \leq 2^{-\lambda}$, with λ denoting the cryptographic security parameter (e.g., 128-bit AES). Thus, data confidentiality holds up to the TEE’s residual side-channel leakage ℓ_{TEE} .

2) *Integrity Argument*: By the measured-boot chain, any modification of the enclave binaries or configuration changes $H(D_i^t)$. Because the CAS signs only verified measurements, the modified enclave fails attestation, and consequently:

$$K_i \notin D_i^t; \text{ hence, } \mathcal{S}_i \text{ is never materialized} \quad (6)$$

Formally, for any modification operator Δ applied by $\mathcal{A}$,

$$\mathbb{P}[\text{Attest}(\mathcal{P}_i \oplus \Delta) = \text{Valid}] = 0. \quad (7)$$

Therefore, $\mathcal{A}$ cannot inject or alter $\theta_i, \nabla_{\theta_i} G_i$, and / or $\mathcal{E}_i$ within an attested enclave instance.

3) *Channel and Leakage Bound*: Let the total leakage channels available to $\mathcal{A}$ be:

$$\mathcal{L} = \mathcal{L}_{\text{CACHE}} \cup \mathcal{L}_{\text{TIMING}} \cup \mathcal{L}_{\text{SPECULATIVE}} \quad (8)$$

For modern enclaves (e.g., Intel SGX, AMD SEV, ARM CCA), mitigations bound observable information leakage by a negligible function $\ell_{\text{TEE}} = f(t, \nu)$, where t is the timing resolution, and ν is the noise entropy. Thus, the adversary’s advantage satisfies:

$$\text{Adv}_{\mathcal{A}}^{\text{TEE}} = \left| \mathbb{P}[\mathcal{A} \rightarrow \mathcal{S}_i] - \frac{1}{2} \right| \leq \ell_{\text{TEE}}. \quad (9)$$

In other words, $\mathcal{A}$ ’s posterior knowledge of $\mathcal{S}_i$ is statistically indistinguishable from random guessing.

4) *Composition with Secure Channel*: Communications between enclave D_i^t and server or other enclaves are authenticated via mutually attested TLS (mTLS) using session key K_i^{SESS} bound to the enclave identity. Hence, even if $\mathcal{A}$ mounts a man-in-the-middle attack within D_i^u , the channel transcript leakage remains bounded by:

$$\mathbb{P}[\text{Forge}_{\mathcal{A}}(K_i^{\text{SESS}})] \leq 2^{-\lambda}. \quad (10)$$

Therefore, no feasible strategy exists for $\mathcal{A}$ to extract or alter $\mathcal{S}_i$ during transmission or at rest.

Next, combining the confidentiality, integrity, and leakage bounds, we have:

$$\mathbb{P}[\mathcal{A} \rightarrow \mathcal{S}_i] \leq \ell_{\text{TEE}} + 2^{-\lambda} \approx \ell_{\text{TEE}}. \quad (11)$$

Hence, under the stated assumptions of a trusted CAS root and a correctly implemented TEE, the adversary’s success probability in obtaining or modifying any element of $\mathcal{S}_i$ is negligible. Hence, proved.

TABLE III
SFL FRAMEWORK GENERALIZES OR OUTPERFORMS FLCERT, CROWDGUARD, AND PPFL IN ROBUSTNESS AND TRUST ENFORCEMENT.

Dimension	SFL [Ours]	FLCERT [?]	CROWDGUARD [?]	PPFL [?]
Primary Defense Locus	Client + Server (end-to-end)	Server (certified robust aggregation)	Server (federated backdoor detection)	Client (TEE isolation) + Server
Adversary Model	Bounded malicious fraction β adaptive / stealthy poisoning, model theft, MITM	Byzantine/poisoning; mostly update-level	Backdoors / poisoning (coordinated)	Host compromise vs. TEE; honest-but-curious
Client-side Input Filtering	Yes; ensemble AEOCL (pre-aggregation)	No	No (post-hoc server detection)	No
Enclave-Protected Training	Yes; model + filter in TEE (measured boot, attestation)	No	No	Yes; model in TEE
Confidentiality / Integrity / Authenticity	All three (attested channels, signed deltas, integrity chain)	Integrity (aggregation certificate); weak on confidentiality / authenticity	Integrity focus (detector); limited confidentiality / authenticity	Confidentiality+ integrity via TEE; limited authenticity
Forensic Visibility (Auditability)	High; provenance logs, trust scores, rollback	Moderate; certified bounds, limited telemetry	Moderate; detector outputs	Low-to-Mod; TEE opacity, limited server insight
Robustness Mechanism	Layered: pre-filter + ZT gate + robust aggregation	Certified robust aggregation	Backdoor detection / triage	Isolation (TEE) + standard FL
Defense Against Adaptive/Covert Attacks	Strong; refreshable ensembles + behavior models	Moderate; bounded by cert assumptions	Moderate; detector evasion possible	Limited; TEE $\neq$ data poisoning defense
Privacy Options (DP / SecAgg)	DP-SGD optional; keeps forensics (no blind SECAGG)	Compatible; may reduce detectability	Compatible; may reduce visibility	Compatible; enclave may reduce DP need
Overhead on Edge	Low-to-Moderate; (lightweight ensembles; enclave-optimized)	Low; (server-heavy)	Low-to-Moderate; (server-side compute)	Moderate; (enclave context + I/O)
Scalability (Heterogeneous IoT)	High; designed for resource-variance	High; server-centric	High; server-centric	Moderate-to-High; TEE / device constraints

TABLE IV

ACCURACY OF DIFFERENT FILTER ENSEMBLES (USING A VARYING NUMBER OF CLASSIFIERS OR m) AGAINST VARIOUS CORRUPTED TRAINING DATA (INCLUDING MNIST, GTSRB, AND LSUN-BEDROOM) CONTAINING 5%-20% POISONED SAMPLES (PREPARED USING *mean-shift perturbations* [?]) PER CLASS, I.E., $\rho \in (0.05, 0.2)$. THIS COMPUTATION IS AVERAGED OVER 80 ACTIVE C_i S AFTER 5000 GLOBAL ITERATIONS.

MNIST							GTSRB						
ρ	← No. of RFs →			← No. of SVMs →			ρ	← No. of RFs →			← No. of SVMs →		
$\downarrow$	$m=1$	$m=3$	$m=5$	$m=1$	$m=3$	$m=5$	$\downarrow$	$m=1$	$m=3$	$m=5$	$m=1$	$m=3$	$m=5$
0	0.000	0.000	0.000	0.000	0.000	0.000	0	0.000	0.000	0.000	0.000	0.000	0.000
0.05	96.091	96.011	93.332	95.712	93.011	92.125	0.05	97.312	95.112	94.019	96.089	95.219	94.001
	-3.717	-1.121	-0.239	-3.921	-1.772	-0.301		-3.092	-1.201	-0.409	-3.091	-1.001	-0.891
	92.374	94.890	93.093	91.791	91.239	91.824		94.229	93.911	93.610	92.998	94.219	93.110
0.10	-2.912	-1.315	-0.911	-3.089	-1.904	-0.921	0.10	-7.219	-5.331	-1.619	-7.998	-5.582	-1.658
	93.179	94.696	92.421	92.623	91.107	91.204		90.102	89.781	92.400	88.091	89.637	92.343
0.15	-10.711	-6.011	-3.089	-11.981	-7.790	-5.101	0.15	-10.18	-9.39	-5.87	-10.91	-9.22	-4.91
	85.38	90.000	90.243	83.731	85.221	87.024		87.141	85.722	88.149	85.179	85.999	89.0911
0.20	-35.091	-15.011	-4.091	-34.981	-16.011	-6.101	0.20	-38.31	-14.31	-9.56	-35.11	-18.89	-8.81
	61.000	81.000	89.241	60.731	77.000	86.024		59.011	80.802	84.459	60.979	76.32	85.191

LSUN-Bedroom							LSUN-Bedroom						
ρ	← No. of RFs →			← No. of SVMs →			ρ	← No. of RFs →			← No. of SVMs →		
$\downarrow$	$m=1$	$m=3$	$m=5$	$m=9$	$m=11$	$m=21$	$\downarrow$	$m=1$	$m=3$	$m=5$	$m=9$	$m=11$	$m=21$
0	0.000	0.000	0.000	0.000	0.000	0.000	0	0.000	0.000	0.000	0.000	0.000	0.000
0.05	98.347	97.134	96.181	96.141	94.118	93.165	0.05	98.211	97.399	97.115	95.482	95.678	94.117
	-3.471	-2.691	-2.582	-1.412	-1.881	-0.813		-3.891	-3.486	-2.856	-2.718	-1.912	-0.881
	94.876	95.443	95.599	94.729	92.237	92.352		94.320	93.913	94.259	92.710	93.766	93.236
0.10	-7.991	-6.682	-6.497	-4.678	-1.991	-1.826	0.10	-8.761	-8.210	-7.391	-5.861	-5.136	-5.021
	90.356	92.452	89.684	91.463	92.127	91.399		89.450	89.189	89.724	89.567	90.544	89.096
0.15	-14.48	-9.57	-7.87	-7.23	-4.01	-3.09	0.15	-18.367	-16.091	-12.231	-12.094	-11.119	-11.998
	83.867	87.564	88.311	88.911	90.108	90.075		79.844	81.308	84.884	83.334	84.559	82.119
0.20	-38.091	-19.183	-15.177	-9.079	-6.157	-6.153	0.20	-37.511	-26.998	-17.091	-17.331	-10.119	-9.335
	60.256	77.951	81.004	87.062	87.961	87.012		60.700	70.401	80.024	78.097	85.559	84.782

TABLE V

ACCURACY OF THE MODELS ON THE PRISTINE BASELINE DATASETS BOTH WITH IID AND NON-IID ASSUMPTIONS. **BLUE** AND **RED** HIGHLIGHTS ARE RESPECTIVELY THE VALUES CORRESPONDING TO THE BEST MODELS FOR IID AND NON-IID ASSUMPTIONS.

On Pristine MNIST: Accuracy 97.0% (Approx.)							On Pristine CIFAR-10: Accuracy 98.0% (Approx.)						
Models	τ	α	U	IID	1-C	5-C	Models	τ	α	U	IID	1-C	5-C
Conv(128, 6, 5)	1	10	4591	321	212	102	Conv(128, 6, 5)+ReLU	1	10	5300	124	142	112
	5	100	4510	31	144	150		5	100	5100	41	190	165
Conv(128, 5, 5)	1	10	4315	179	223	70	Conv(128, 5, 5)+ReLU	1	10	5120	109	161	113
	5	100	5430	473	145	67		5	100	5000	116	72	38
Conv(64, 5, 5)	1	10	5430	175	112	69	Conv(64, 5, 5)+ReLU	1	10	5000	120	101	131
	5	100	4560	155	241	72		5	100	5000	45	103	112
FC(300)+ReLU	1	10	4150	212	134	112	FC(300)+ReLU	1	10	5050	111	142	121
	5	100	4280	200	64	53		5	100	5280	131	109	145
FC(128)+ReLU	1	10	4200	193	122	113	FC(128)+ReLU	1	10	5200	170	169	180
	5	100	3500	39	187	127		5	100	5500	117	104	182
On Pristine LSUN-Bedroom: Accuracy 98.0% (Approx.)							On Pristine GTSRB: Accuracy 98.0% (Approx.)						
Conv(128, 6, 6)+ReLU	1	10	6900	296	212	134	Conv(128, 6, 6)	1	10	5141	110	103	132
	5	100	3810	25	100	121		5	100	5310	21	35	39
Conv(128, 5, 5)+ReLU	1	10	5715	379	221	159	Conv(128, 5, 5)	1	10	5315	119	112	110
	5	100	7320	213	192	202		5	100	5130	151	172	192
Conv(64, 5, 5)+ReLU	1	10	5912	148	161	187	Conv(64, 5, 5)	1	10	5150	141	109	123
	5	100	6123	173	170	152		5	100	5160	191	132	123
FC(300)+ReLU	1	10	6150	217	321	302	FC(300)+ReLU	1	10	5150	412	201	218
	5	100	6280	311	213	219		5	100	5280	143	138	182
FC(128)+ReLU	1	10	1200	252	191	129	FC(128)+ReLU	1	10	5200	158	206	209
	5	100	3500	87	34	29		5	100	5500	137	123	141

TABLE VI

ACCURACY OF BASELINE MODELS ON POISONED DATASETS UNDER IID AND NON-IID ASSUMPTIONS WITH 9 CORRUPTED CLIENTS. **BLUE** AND **RED** INDICATE THE BEST MODELS FOR IID AND NON-IID, RESPECTIVELY.

On Corrupted MNIST with $\epsilon = 0.05$, $\gamma = 0.20$								On Corrupted CIFAR-10 with $\epsilon = 0.05$, $\gamma = 0.20$							
Models	τ	γ	U	IID ¹	IID ²	1-C ¹	5-C ²	Models	τ	γ	U	IID ¹	IID ²	1-C ¹	5-C ²
Conv(128, 6, 5)	1	10	5100	110	101	117	119	Conv(128, 6, 5)	1	10	6500	89	95	110	128
	5	100	5100	49	58	115	109		5	100	6500	65	69	110	113
Conv(128, 5, 5)	1	10	5100	117	132	119	112	Conv(128, 5, 5)	1	10	6500	102	119	165	116
	5	100	5100	111	101	94	99		5	100	6500	135	115	82	73
Conv(64, 5, 5)	1	10	5100	135	133	142	143	Conv(64, 5, 5)	1	10	6500	189	157	112	119
	5	100	5100	130	102	105	108		5	100	6500	112	111	190	181
FC(300)	1	10	5400	132	145	152	115	FC(300)	1	10	6500	110	119	123	113
	5	100	5400	111	115	69	71		5	100	6500	182	113	109	134
FC(128)	1	10	5400	145	141	121	119	FC(128)	1	10	6500	109	120	119	140
	5	100	5400	109	105	114	116		5	100	6500	109	102	111	131
On Corrupted LSUN-Bedroom with $\epsilon = 0.05$, $\gamma = 0.20$								On Corrupted GTSRB with $\epsilon = 0.05$, $\gamma = 0.20$							
Conv(128, 6, 5)	1	10	6500	182	198	201	206	Conv(128, 6, 5)	1	10	6000	114	129	98	109
	5	100	5000	84	89	153	139		5	100	6000	56	60	69	60
Conv(128, 5, 5)	1	10	5000	113	120	170	168	Conv(128, 5, 5)	1	10	6500	128	148	107	110
	5	100	5000	131	155	202	201		5	100	6500	111	109	97	109
Conv(64, 5, 5)	1	10	5000	105	100	113	130	Conv(64, 5, 5)	1	10	6500	101	162	165	189
	5	100	5000	113	119	202	211		5	100	6500	97	101	110	119
FC(300)	1	10	5000	120	101	119	142	FC(300)	1	10	6500	119	163	178	102
	5	100	5000	182	142	102	112		5	100	6500	129	119	156	152
FC(128)	1	10	5000	98	103	112	149	FC(128)	1	10	6500	129	112	109	134
	5	100	4800	108	113	49	52		5	100	6500	119	130	156	104
+ ReLU								+ ReLU							